

04 Understanding Table Tags

Table structure · colspan and rowspan · Accessibility

The complete structure of a table

```
<table>
  <caption>Semester 1 Result</caption>
  <colgroup><col span="1"></colgroup>
  <thead>
    <tr><th scope="col">Name</th><th scope="col">Marks</th></tr>
  </thead>
  <tbody>
    <tr><th scope="row">Aman</th><td>88</td></tr>
  </tbody>
  <tfoot>
    <tr><td>Average</td><td>88</td></tr>
  </tfoot>
</table>
```

Tag	Purpose
<code><table></code>	The container for the entire table.
<code><caption></code>	The title of the table. It must be the first child of <code><table></code> .
<code><thead></code> , <code><tbody></code> , <code><tfoot></code>	Logical grouping of header, body and footer rows. When a long table is printed, the browser can repeat the <code><thead></code> on every page.
<code><tr></code>	A table row, that is, one horizontal line of cells.
<code><th></code>	A header cell, bold and centred by default. It should always carry <code>scope="col"</code> or <code>scope="row"</code> .
<code><td></code>	A standard data cell.
<code><colgroup></code> , <code><col></code>	Applies width or styling to entire columns at once.

colspan and rowspan

- `colspan="2"` makes a cell span two columns horizontally.
- `rowspan="3"` makes a cell span three rows vertically.
- The rule that matters most:** when a cell spans downwards, the following rows must contain **that many fewer** cells. Otherwise the grid breaks and an extra column appears.

```
<tr><td rowspan="2">Aman</td><td>Mathematics</td></tr>
<tr><td>Science</td></tr>  <!-- only one cell, because the cell above already occupies this column -->
```

Accessibility and best practices

- Never use tables for page layout.** That was a practice of the 1990s. Layout today belongs to CSS Flexbox and Grid, and layout tables destroy the reading order for screen readers.
- The `scope` attribute tells assistive technology which cells a header governs. Without it, tables with two levels of headers become impossible to interpret.
- A `<caption>` supplies context that benefits sighted and non-sighted users alike.
- Presentational attributes such as `border` , `cellpadding` , `cellspacing` and `bgcolor` are obsolete in HTML5. Use CSS instead, beginning with `border-collapse: collapse`.
- On small screens, wrap a wide table in a container with `overflow-x: auto` rather than removing columns, which would damage the data.

COMMON MISTAKES

Writing `<td>` outside a `<tr>` · forgetting to remove a cell after using `rowspan` · using `<td>` where a row header is required · omitting the caption · styling with obsolete HTML attributes.

QUICK RECAP

- Tables are for tabular data only, never for layout.
- `colspan` merges horizontally, `rowspan` merges vertically, and both change the cell count of the rows they affect.
- `caption`, `th` and `scope` together make a table understandable to assistive technology.